

Project Execution README

MLOPS ASSIGNMENT 2 — MODEL TRAINING & EVALUATION TRACKING

1. Overview

This repository contains the setup, pipeline implementation, and evaluation metrics for MLOps Assignment 2. The core implementation was adapted from an established baseline pipeline and optimized to execute smoothly within a cloud-hosted environment (Kaggle). The workflow integrates external Hugging Face architectures alongside Weights & Biases (W&B) to maintain granular tracking of model fine-tuning checkpoints, systemic losses, and accuracy deltas across epochs.

2. Setup & Installation Instructions

To mirror or reproduce the experimental run conducted in the Kaggle environment, execute the structural pipeline setup described below:

Environment Initialization

1. Base dependencies must be updated and satisfied using the dedicated environment manifest file. Build the required stack sequentially inside your notebook container or terminal profile:

```
pip install -r requirements.txt
```

Authentication Configuration

2. Prior to initializing the core runtime script, ensure your environment exposes the necessary security tokens to interface with remote endpoints:

- **Hugging Face API Token:** Required to authenticate, fetch architectural weights, and pull base models or secure endpoints seamlessly.
- **W&B API Token:** Essential for establishing connection with the workspace, enabling direct synchronization of loss curves and run performance variables.

Note on Execution Flow: The entire process — tracking setup, package parsing, model ingestion, parameter adjustments, evaluation passes, and live telemetry shipping — is designed to run end-to-end within a continuous, single-pass pipeline execution framework once the API keys are injected into the kernel secrets profile.

3. Execution Workflow

The pipeline was migrated and stabilized through the following milestones:

1. **Source Extraction:** Ingested core baseline components from the foundational source notebook repository.
2. **Environment Porting:** Imported structural blocks directly into a target Kaggle Notebook sandbox environment.
3. **Code Auditing & Optimization:** Audited source syntax structures and adjusted configurations where environment paths or storage allocation paradigms drifted. Added structural integrations to wire platform tokens dynamically.
4. **End-to-End Execution:** Triggered standard dependency installation and launched the end-to-end pipeline in a single sequential execution block to guarantee state continuity.

4. Experimental Evaluation & Results

Model metrics were validated dynamically against evaluation benchmarks upon training termination. Performance characteristics yielded the following metrics:

Evaluation Metric	Observed Output / Final Value
eval/accuracy	0.5675 (56.75%)
eval/f1	0.58084
eval/loss	2.66998
final/accuracy	0.5675 (56.75%)
final/f1	0.58084

5. Project Environment & Artifact Registry

Live model runs, pipeline instances, baseline definitions, and workspace telemetry are distributed across the following cloud registries:

Kaggle Notebook: <https://www.kaggle.com/code/sudebghoshg25ait2113/notebook87162508f5>
Hugging Face: <https://huggingface.co/sudebghosh>
W&B Dashboard: <https://wandb.ai/g25ait2113-iitj/mlops-assignment2?nw=nwuserg25ait2113>
Source Colab: <https://colab.research.google.com/drive/15yJsCtRu4kgqCLT44Tjhs3SF0T5GITqC?usp=sharing>